

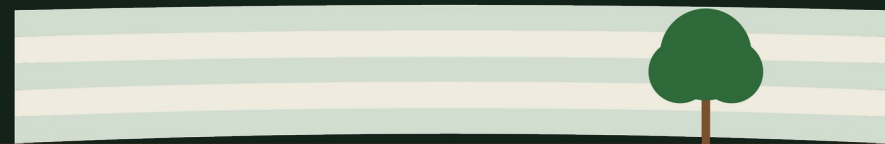
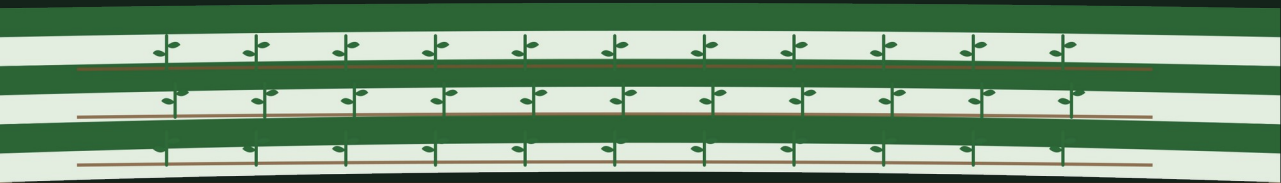
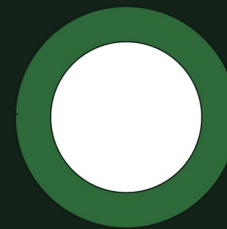


DEALER TRAINING · SEPTEMBER 2026

V-Mag Essential

Magnesium 8.5% with boron 1%, a powder dissolved into the irrigation while the canopy is being built.

For VAN dealers and the field team, written so a dealer can repeat it to a grower. Lahore.



What a dealer says in 30 seconds.

V-Mag Essential is magnesium 8.5% with boron 1%, a powder that dissolves for the water: 425 g of magnesium and 50 g of boron in a 5 kg bag. Magnesium sits at the centre of every chlorophyll molecule. No magnesium, no green, no sugar. It is pushed out of the root by too much potash and by too much calcium, so a programme that raises potash on calcareous ground often shows magnesium shortage the same season: the older leaves go yellow between the veins while the veins stay green. The Punjab survey does not test magnesium, so no district figure exists; the leaf tells the grower. The programmes run it through the water at early growth and grand growth, ½ bag each on wheat, 1 bag each on maize and potato, while the canopy is being built. Dissolve it in the drum, release it across the irrigation.

The 4 things it must contain

1

The bag: 5 kg, Mg 8.5% · B 1%, powder. 425 g of magnesium and 50 g of boron in every bag. No PSQCA licence; a conformance report. Ask VAN.

2

The nutrient: the centre of chlorophyll. Pushed out by too much potash and calcium, which is the Pakistani programme on Pakistani ground.

3

The sign: older leaves yellow between green veins, striped, from the bottom up. Not the whole leaf, which is nitrogen.

4

The method: dissolved in the drum, through the water at early and grand growth, at the row the plan prints. Never a guessed dose.

Say it in the grower's own words. Every figure in these 30 seconds is on the bag, in the published crop programme, or in the named evidence on the slides that follow.

THE BAG

What the bag says, and what it is registered as.

5 kg

the bag. 425 g of magnesium and 50 g of boron in it, from the declared analysis. ½ bag per application on wheat; 1 bag on maize, potato, tomato and citrus.

Powder

a mineral powder that dissolves for fertigation, as VAN’s product data says. The citrus rows also spray it, 400 g.

Mg 8.5%

magnesium, by weight: 85 g in every kilogram, 425 g in the bag. The analysis VAN confirmed on 23 July 2026.

Not tested

magnesium is not among the 13 determinations in the Punjab survey of 770,160 samples, so no district figure exists. The leaf is the test.

B 1%

boron, 10 g in every kilogram, 50 g in the bag. Small and exact, because the gap between enough boron and too much is narrow.

19

published programmes name V-Mag Essential, 32 rows, fertigation on 30 of them.



Registered as	Pakistan Standard	Manufacturing licence
V-Mag Essential	No Pakistan Standard applies (n/a)	Conformance report; registration: ask VAN

No expiry date is printed in this deck. The licence number is what a dealer quotes.

A GROWER'S QUESTION

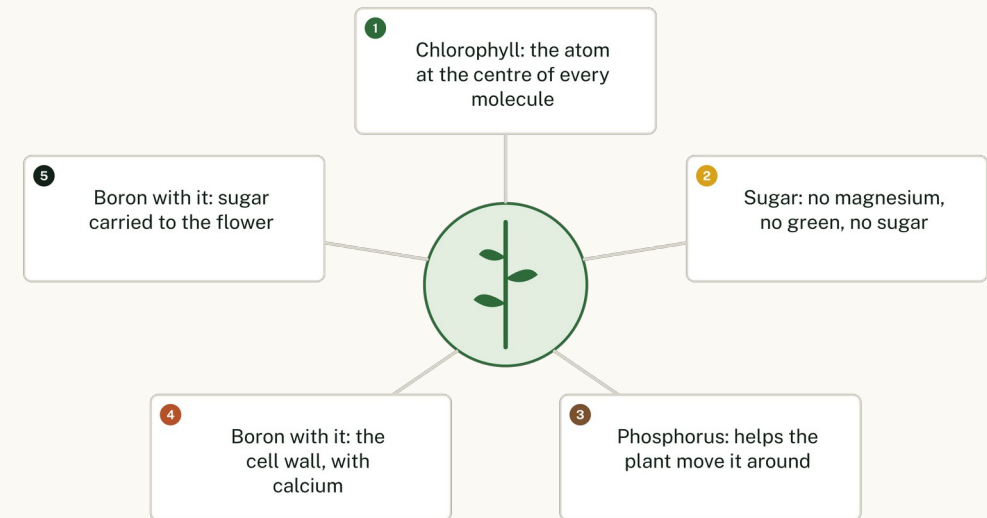
“I gave the crop its potash properly this year, and now the old leaves have gone yellow in stripes between the veins, from the bottom up. Urea did not fix it. What is it?”

The grower did the right thing and got a new problem. Potash and magnesium compete at the root, so the better his potash programme, the more likely magnesium is the nutrient that ran short. The answer is 3 ideas: what magnesium does, why the sign shows on the old leaves and not the new, and why the fix goes through the water while the canopy is being built.

The 3 answers on the next 3 slides, 1 idea each.

Magnesium is the centre of the green.

Magnesium sits at the centre of every chlorophyll molecule: without it the leaf cannot make green, and without green it cannot make sugar. It also helps the plant move phosphorus around, so the energy phosphorus carries is spent through magnesium. A crop short of it is standing in light it cannot use. The boron in the bag does its own job beside it: it builds the cell wall with calcium and carries sugar to the flower.



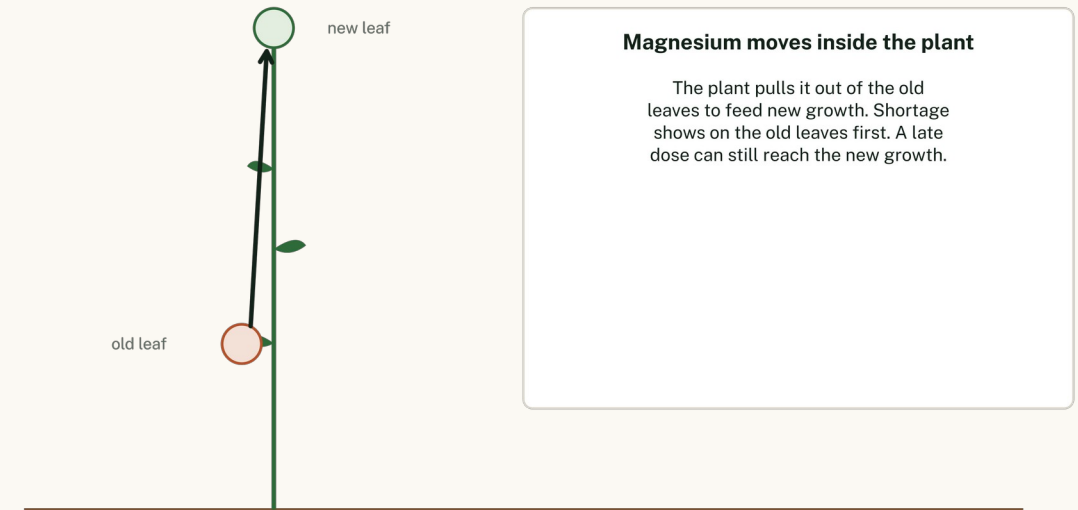
The jobs of magnesium and boron, from the site's 17-nutrients page. A picture of the idea, not a measurement.

Say it like this: "Magnesium is the atom in the middle of the green.
A crop short of it is standing in sunlight it cannot use."

It moves inside the plant, so the old leaves show it first.

Magnesium is mobile in the plant. When the supply runs short, the plant strips it out of the old leaves to feed the new growth, so the sign appears low on the plant and moves up: yellow between the veins while the veins stay green, a striped or marbled look, starting at the margin. That is how a dealer tells it from nitrogen, which yellows the whole old leaf from the tip back, and from iron, which yellows the youngest leaves. By the time the stripes show, the canopy has been short for a while.

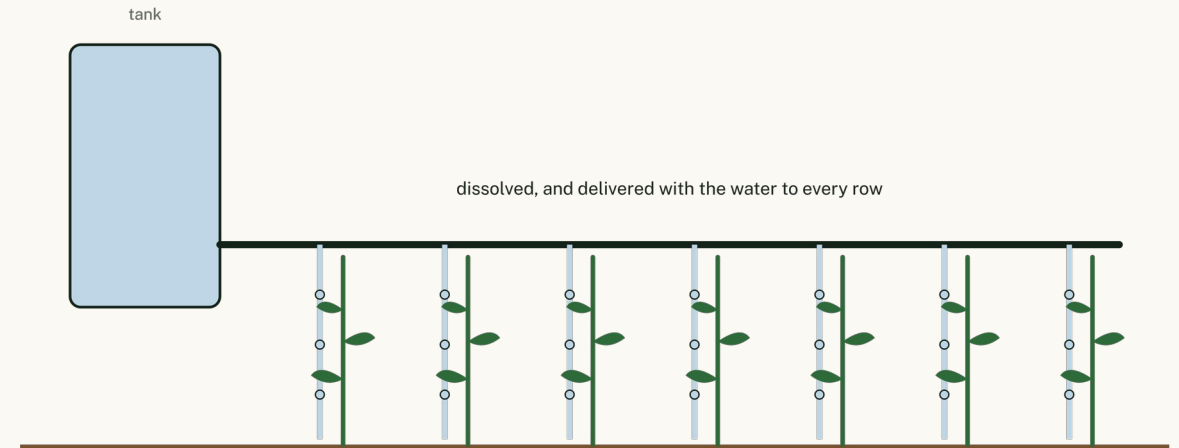
Say it like this: “Stripes on the old leaves, veins still green, from the bottom up: that is magnesium. Whole leaf yellow from the tip: that is nitrogen.”



A mobile nutrient: the plant robs the old leaf to feed the new one. General plant physiology, stated as such.

Potash and calcium push it out. The fix goes through the water.

Magnesium is pushed out of the root by too much potash and by too much calcium. Pakistani ground is full of calcium as lime, and a good programme raises potash, so a crop can run short of magnesium in the season the potash goes right. A soil that has never been tested for magnesium cannot warn the grower; the Punjab survey does not measure it. So the programmes carry it through the water, dissolved, at early growth and grand growth while the canopy is being built, and on onion the row runs it in the same water as the Vital Potash it balances.



The fertigation line: a solution released across the irrigation. No soil figure for magnesium exists on the record, and none is drawn.

Say it like this: “The better your potash, the more the crop needs its magnesium. Put it through the water while the canopy is growing.”

THE EVIDENCE

What is verified for V-Mag Essential, and what is not.

No trial of V-Mag Essential is on VAN's record, and it carries no PSQCA manufacturing licence. What a dealer may state as fact is the bag, the conformance report, the programme rows and the soil record's silence on magnesium, each as what it is.

Item	What it says	How a dealer may use it
The bag	V-Mag Essential, Mg 8.5% · B 1%, mineral powder, 5 kg. The analysis confirmed by VAN on 23 July 2026.	As fact. Grams from the declared analysis.
Registration	No Pakistan Standard applies; the master record holds a conformance report, and the site says so: 1 of 3 VAN brands without a PSQCA manufacturing licence, each for a stated reason.	As the site prints it. "Registration: ask VAN." Never "licensed" and never a licence number.
The programme rows	32 rows on 19 programmes: ½ bag twice on wheat, chili and sunflower; 1 bag twice on maize, potato, tomato, banana, watermelon; 1 bag once on sugarcane, ratoon, basmati and hybrid rice; 2.5 kg twice on onion with Vital Potash; citrus 1 bag fertigated and 400 g sprayed twice.	As the published rate. Verbatim.
The soil record (Punjab, 770,160 samples)	Magnesium is not among the 13 determinations. No district figure exists for it. 69.1% of samples read above pH 8; the ground is calcareous. 51.7% read low in boron.	As fact. The absence is the point: nobody has measured magnesium, so the leaf is the test. Never as product performance.
The chemistry (textbook)	Magnesium is the centre of chlorophyll; it is mobile in the plant; too much potash and too much calcium push it out at the root.	As general plant nutrition, from the site's 17-nutrients page, not a VAN measurement.
Not on file	A V-Mag Essential trial; a yield or quality figure; a Pakistani magnesium deficiency figure; the salt the magnesium comes as; a soil-application rate; a leaf-test threshold.	Not said until named.

Sources: the bag; VAN's published crop programmes on van.com.pk; the farmer price sheet, Aug 2026; the regulatory table on van.com.pk; the Punjab soil record, 770,160 samples, and its list of determinations; VAN's 17-nutrients page; VAN's own product data.

The bag in grams, and the wheat season, in the grower’s own back-of-envelope.

Price from the VAN farmer price sheet, Aug 2026, delivered farmer price. Magnesium and boron from the declared analysis. The wheat programme puts ½ bag on at early growth and ½ at grand growth, through the water.

V-Mag Essential, Mg 8.5 · B 1	Weight	Magnesium	Boron	Rupees	Which rows
1 bag	5 kg	425 g	50 g	1,350	maize, potato, tomato, sugarcane, rice, citrus
½ bag	2.5 kg	212.5 g	25 g	675	wheat, chili, sunflower; onion prints 2.5 kg
400 g, sprayed	400 g	34 g	4 g	108	citrus, at germination and grand growth
Wheat, ½ bag at early growth and ½ at grand growth	5 kg	425 g	50 g	1,350	1 bag for the season

On wheat the grower spends 1,350 rupees an acre across the season to put 425 g of magnesium and 50 g of boron through the water at the 2 stages the canopy is built. That is the counter arithmetic. What the green does with it is not a number VAN prints for this product.

V-Mag Essential 5 kg PKR 1,350: the farmer price sheet, Aug 2026. 425 g = 5,000 × 0.085; 50 g = 5,000 × 0.01; 34 g = 400 × 0.085. Rates: the crop programmes on van.com.pk.



What the programme puts on, per acre, per season: the field crops.

Bags per acre from the published programmes; magnesium from the declared analysis; rupees at the VAN farmer price sheet, Aug 2026.

Crop	Stage and method, as printed	Bags	Magnesium	Rupees per acre
Wheat	Early growth ½, grand growth ½, fertigation	1	425 g	1,350
Maize	Early growth 1, grand growth 1, fertigation	2	850 g	2,700
Sunflower	Early growth ½, grand growth ½, fertigation	1	425 g	1,350
Sugarcane and ratoon	Grand growth 1, fertigation	1	425 g	1,350
Basmati and hybrid rice	Grain formation and maturity 1, fertigation	1	425 g	1,350

Most field crops take 1 bag a season in 2 halves, at the 2 stages the canopy is built. Maize takes 2. The whole line runs through the water. That is the counter arithmetic; what the green does with it is not a number VAN prints.

VAN's published crop programmes on van.com.pk, rows as printed. Rupees = bags × 1,350, farmer price sheet, Aug 2026.

What the programme puts on, per acre, per season: vegetables and orchards.

Bags per acre from the published programmes; magnesium from the declared analysis; rupees at the VAN farmer price sheet, Aug 2026.

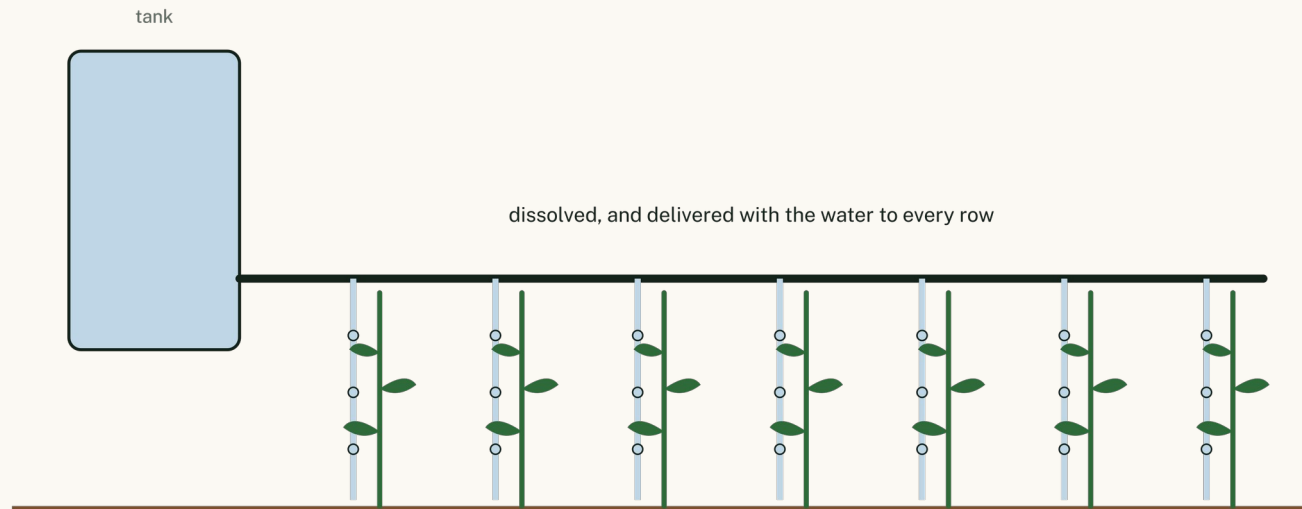
Crop	Stage and method, as printed	Bags	Magnesium	Rupees per acre
Potato	Early growth 1, grand growth 1, fertigation	2	850 g	2,700
Tomato	Germination 1, grand growth 1, fertigation	2	850 g	2,700
Chili	Early growth ½, after picking ½, fertigation	1	425 g	1,350
Onion	2.5 kg at 40 and 50 days, fertigation with Vital Potash	1	425 g	1,350
Citrus	1 bag fertigated at germination; 400 g sprayed at germination and grand growth	1 + 800 g	493 g	1,566

The vegetables take 2 bags. Onion is the 1 row that names the tank partner: the magnesium goes in with the Vital Potash it balances. Citrus is the 1 crop that also sprays it, 400 g at 2 stages.

VAN’s published crop programmes on van.com.pk, rows as printed. Rupees = bags × 1,350, farmer price sheet, Aug 2026; 800 g of spray = 0.16 bag = 216 rupees. 493 g = 425 + 68.

HOW IT GOES ON

Dissolve it in the drum. Release it across the irrigation. Measure the row's amount.



Weigh the row's amount

½ bag is 2.5 kg; onion prints 2.5 kg; citrus sprays 400 g. Weigh it. A guessed dose is the commonest error with a powder, and the boron in the bag is the reason not to guess upward: 50 g in a bag is exact on purpose.

Dissolve, and keep stirring until it is clear

Into a drum of water at the head of the watercourse, or the drip tank, stirred until it is fully in solution. Undissolved powder settles and blocks the tap or the emitter. VAN's product data draws a 200 L drum because that is what stands at the head of most Punjabi watercourses: scale the water, never the product.

Meter it in, with the water already flowing

Open the tap so the drum empties across the whole irrigation, never in 1 slug, so the whole acre gets it in the wetted root zone. At early growth and grand growth on most rows; with the Vital Potash on onion. Citrus alone also sprays 400 g, at the plan's 2 stages. No soil-spreading rate is printed, because no programme row gives one.

“1,350 for 5 kg of magnesium salt? The chemist sells Epsom salt by the kilo.”

The dealer says, out loud

Write both on your paper; VAN prints no price for a general salt. Then write what the bag is: 425 g of magnesium with 50 g of boron already measured into it, made to dissolve clear for the drum, at the rate the plan prints, ½ bag on wheat per application. You are buying the magnesium, the boron beside it in an exact small amount, and a powder that goes through the tap. What the green does with it is not a number VAN prints.

The fact behind it

- V-Mag Essential 5 kg is 1,350 rupees; ½ bag on wheat is 675 (farmer price sheet, Aug 2026; the wheat programme on van.com.pk).
- 425 g of magnesium and 50 g of boron per bag (the declared analysis on the bag).
- A powder that dissolves for fertigation (VAN's product data).
- Wheat: ½ bag at early growth and ½ at grand growth, through the water (the wheat programme on van.com.pk).

Never say: “It gives you X more maunds” or a rupees-per-acre return. No yield figure exists for the product. No cost per gram against the general salt.

“My soil report says nothing about magnesium. So my soil is fine?”

The dealer says, out loud

It says nothing because nobody measured it. Magnesium is not one of the 13 things the Punjab survey tests, so no district in the province has a figure, and a silent report is not a clean report. What tells you is the leaf: old leaves yellow between the veins, in stripes, from the bottom up, in the season your potash went right. If you want the number, ask for a soil or leaf test that includes magnesium.

The fact behind it

- The Punjab survey of 770,160 samples runs 13 determinations; magnesium is not among them (the Punjab soil record).
- A magnesium-short crop shows older leaves yellow between the veins, veins green, starting low and moving up (VAN's 17-nutrients page).
- Nitrogen shortage yellows the whole old leaf from the tip; iron shortage yellows the youngest leaves (VAN's 17-nutrients page).

Never say: “X% of Pakistani soils are short of magnesium.” No such figure is on VAN's record.

“I already put potash on. Will this fight it?”

The dealer says, out loud

The other way round: the potash is why you need it. Too much potash at the root pushes magnesium out, and too much calcium does the same, and Pakistani ground is full of calcium as lime. So the plan doses the 3 together, never 1 alone. The onion row even puts the magnesium in the same water as the Vital Potash. Keep the potash; add the magnesium at the rows the plan prints.

The fact behind it

- Too much potash pushes out magnesium and calcium, so the 3 are dosed together, never 1 alone (VAN's 17-nutrients page).
- Pakistani soil is full of calcium as lime; 69.1% of Punjab samples read above pH 8 (VAN's 17-nutrients page; the Punjab soil record).
- Onion: 2.5 kg of V-Mag Essential in the same fertigation as Vital Potash, at 40 and 50 days (the onion programme on van.com.pk).

Never say: “Skip the potash and the magnesium will be fine”, or the reverse.

“The old leaves are already striped. Will this green them again?”

The dealer says, out loud

The new growth is what it feeds. Magnesium is mobile, so the plant already robbed those old leaves to feed the top, and a leaf that has yellowed between the veins does not usually go back. Put the row's amount through the water now so the canopy still being built gets its magnesium, and plan the early-growth application next season, before the stripes. VAN prints no recovery figure and no day count.

The fact behind it

- Magnesium moves inside the plant: the old leaves are stripped to feed new growth, so the sign starts low and moves up (VAN's 17-nutrients page).
- Wheat, maize and potato take it at early growth and again at grand growth (the crop programmes on van.com.pk).
- No recovery or yield figure exists for V-Mag Essential.

Never say: “It greens the old leaves in a week” or “it recovers the yield”.

“Why is there boron in a magnesium bag?”

The dealer says, out loud

Because the 2 travel together in the plan. Boron builds the cell wall with calcium and carries sugar to the flower; magnesium makes the green that makes the sugar. 51.7% of Punjab samples read low in boron, and boron is the nutrient where the dose has to be small and exact, so 50 g in a 5 kg bag, spread through the water, is a safe way to carry it. Keep to the row: boron adds up across products, and more of it is worse.

The fact behind it

- Boron builds the cell wall with calcium and carries sugar to the flower; the gap between enough and too much is narrow (VAN's 17-nutrients page).
- 51.7% of Punjab samples read below 0.5 ppm boron (the Punjab soil record).
- 50 g of boron per bag: $5,000 \text{ g} \times 0.01$; a wheat season is 1 bag, 50 g (the declared analysis; the wheat programme on van.com.pk).
- Boron adds up across products; the plan's rows are the whole dose (textbook agronomy).

Never say: “Boron cannot harm the crop” or an extra bottle of boron on a crop already getting it here and on the coated potash.

“Can I spread it on the soil instead of dissolving it?”

The dealer says, out loud

Every programme row runs it through the water, and 2 citrus rows spray it; no row spreads it on the soil, so I have no soil rate to give you. It is a powder made to dissolve. Put it in the drum, stir it clear, and meter it across the irrigation. If you cannot fertigate that field, ask VAN before you spread it.

The fact behind it

- Fertigation on 30 of the 32 rows; the other 2 are citrus sprays; no row spreads it on the soil (the crop programmes on van.com.pk).
- Fertigation is the route VAN names for the product (the V-Mag Essential page on van.com.pk).
- A powder that dissolves for fertigation (VAN's product data).

Never say: A soil-spreading rate the plan does not print.

“Which stage, and how much?”

The dealer says, out loud

What the plan prints, through the water. Wheat, chili and sunflower: ½ bag at early growth and ½ at grand growth (chili’s second half after picking). Maize, potato and tomato: 1 bag at each of 2 stages. Sugarcane and ratoon: 1 bag at grand growth. Rice: 1 bag at grain formation. Onion: 2.5 kg at 40 and 50 days with the Vital Potash. Citrus: 1 bag through the water and 400 g sprayed at germination and grand growth. Not at land preparation: no row puts it there.

The fact behind it

- 19 programmes name V-Mag Essential, 32 rows (the crop programmes on van.com.pk).
- ½ bag twice on wheat, chili, sunflower; 1 bag twice on maize, potato, tomato, banana, watermelon; 1 bag once on sugarcane, ratoon and rice (the same programmes).
- The stage VAN names for the product is mid-life; no row is at land preparation (the V-Mag Essential page and the programmes on van.com.pk).

Never say: “More bags, more green”, or a rate for a crop with no row.

“Is it registered? What is the licence?”

The dealer says, out loud

V-Mag Essential carries a conformance report, and no PSQCA manufacturing licence, because no Pakistan Standard applies to it; the site says so, as 1 of 3 VAN brands in that position, each for a stated reason. I will not quote you a licence number, because there is none to quote. Every batch is still tested at VAN Lab, PNAC LAB 336, before it ships, and the batch certificate is free on request. For the registration position, ask VAN.

The fact behind it

- V-Mag Essential: no Pakistan Standard applies; it carries a conformance report (the regulatory table on van.com.pk).
- 20 of VAN's 23 brands hold a live PSQCA manufacturing licence; the other 3 each have a stated reason (the regulatory page on van.com.pk).
- Every batch is released through VAN Lab, PNAC LAB 336, ISO/IEC 17025:2017 (VAN's quality page).

Never say: “It is licensed”, a licence number, or “it does not need registration”.

V-Mag Essential in the published programmes, per acre.

Crop	Stage	Rate per acre	Method	Pack
Wheat	Early Growth	½ bag	Fertigation	5 kg
Wheat	Grand Growth	½ bag	Fertigation	5 kg
Maize (Corn)	Early Growth	1 bag	Fertigation	5 kg
Maize (Corn)	Grand Growth	1 bag	Fertigation	5 kg
Basmati Rice	Grain Formation & Maturity	1 bag	Fertigation	5 kg
Hybrid Rice	Maturity	1 bag	Fertigation	5 kg
Sugarcane, February planting	Grand Growth	1 bag	Fertigation	5 kg
Ratoon Sugarcane	Grand Growth	1 bag	Fertigation	5 kg
Potato	Early Growth	1 bag	Fertigation	5 kg
Potato	Grand Growth	1 bag	Fertigation	5 kg
Tomato	Germination	1 bag	Fertigation	5 kg

Rows as published in van.com.pk's crop programmes, 12 of the 19 programmes that name V-Mag Essential. Fertigation on every row here except the 2 citrus sprays; onion prints the tank partner. A rate is per acre; the crop page scales it to the field.

V-Mag Essential in the published programmes, continued.

Crop	Stage	Rate per acre	Method	Pack
Tomato	Grand Growth	1 bag	Fertigation	5 kg
Chili	Early Growth	½ bag	Fertigation	5 kg
Chili	After Picking	½ bag	Fertigation	5 kg
Onion	Grand Growth (40 DOT)	2.5 kg	Fertigation with Vital Potash	5 kg
Onion	Maturity (50 DOT)	2.5 kg	Fertigation with Vital Potash	5 kg
Citrus, 5 years and over	Germination	1 bag	Fertigation	5 kg
Citrus, 5 years and over	Germination	400 g	Spray	5 kg
Citrus, 5 years and over	Grand Growth	400 g	Spray	5 kg
Sunflower	Early Growth	½ bag	Fertigation	5 kg
Sunflower	Grand Growth	½ bag	Fertigation	5 kg

Rows as published in van.com.pk's crop programmes, 12 of the 19 programmes that name V-Mag Essential. Fertigation on every row here except the 2 citrus sprays; onion prints the tank partner. A rate is per acre; the crop page scales it to the field.

WHAT VAN WILL NOT CLAIM FOR IT

A dealer who says any of these is speaking for himself, not for VAN.

1 A yield, quality or greening figure. No V-Mag Essential trial exists; the potash-and-magnesium argument is mechanism only.

2 A Pakistani magnesium deficiency percentage. Nobody has measured it in the Punjab survey, and VAN does not invent the figure the survey lacks.

3 That it is licensed, or a licence number. It carries a conformance report; the registration position is VAN's to state.

4 A soil-spreading rate, a spray water volume, or a leaf-test threshold the plan does not print.

5 What salt the magnesium comes as, or that it is “organic” or “mineral” in any sense beyond a powder that dissolves, until VAN states it.

6 That the boron in the bag is the crop's boron, or that a grower may add more boron on top without looking at the plan.

Every batch is tested. The bag carries the proof.

1 · The batch number

It is printed on the bag. Read it to the grower and show him where it is. Every bag of every batch carries one.

2 · The certificate of analysis

For any batch, free on request. Send the batch number off the bag to kisan@van.com.pk or the helpline, and the certificate for that batch comes back. No sample to post.

3 · The laboratory

VAN Lab, PNAC LAB 336, accredited to ISO/IEC 17025:2017. Every batch is released through it before it ships. The certificate carries the laboratory's number.

What a certificate for V-Mag Essential shows

The declared analysis tested for the batch: magnesium and boron against Mg 8.5% · B 1%, with the batch number on the bag. No Pakistan Standard or licence number applies; the master record holds a conformance report. No certificate for a V-Mag Essential batch was on file for this deck, so no batch reading is printed here; the grower asks for his own batch's certificate.

The lines a dealer says in Urdu.

The first line is the promise word from the VAN brand master. The other 3 are the deck's key sentences, translated in a first pass and not yet reviewed by VAN. Say them, and send corrections to partner@van.com.pk.

Magnesium. VAN has no promise word for it; a plain line.

Urdu first pass, unreviewed

میگنیشیم ہریالی کا مرکز ہے

Stripes on the old leaves, veins still green, from the bottom up:
that is magnesium.

Urdu first pass, unreviewed

پرانے پتوں پر دھاریاں، رگیں ہری، نیچے سے اوپر: یہ میگنیشیم ہے

The better your potash, the more the crop needs its magnesium.

Urdu first pass, unreviewed

پوٹاش جتنی اچھی، فصل کو میگنیشیم کی اتنی ضرورت

Dissolve it in the drum and release it across the whole irrigation.

Urdu first pass, unreviewed

ڈرم میں گھولیں اور پورے پانی کے ساتھ آہستہ چھوڑیں

Write to the address for what you need.

If you grow

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch, a dealer near you.

If you sell

partner@van.com.pk

Dealers, distributors and the trade. Stock, training, corrections to this deck.

Everything else

info@van.com.pk

Supply, the laboratory, a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore. 1 helpline, 3 addresses. No group broadcasts: a grower who writes gets 1 reply.

V-Mag Essential: magnesium and boron through the water, while the canopy is being built. Weigh the row's amount; dissolve it clear.

